Helicobacter pylori Infectious Disease — Comprehensive Disease Characterization

Disease: Helicobacter pylori Infectious Disease **MONDO ID:** MONDO:0006781 **Category:** Infectious Disease **Causative organism:** *Helicobacter pylori* (NCBI Taxon 210) **Report type:** Literature-based disease-level synthesis (no primary patient data). Evidence sources labeled *human clinical*, *model organism*, or *in vitro*.

Summary

Helicobacter pylori infectious disease is a lifelong chronic bacterial infection of the human gastric mucosa caused by the urease-producing, microaerophilic, spiral Gram-negative bacterium *H. pylori*. It is one of the most common chronic infections of humankind: a Bayesian meta-analysis of 181 studies estimated global population prevalence at 36.3% in 2023, down modestly from 43.1% in 1990 (PMID: 42757050). Infection is almost always acquired in childhood via oral–oral, gastric–oral, and fecal–oral routes, and if untreated persists for the life of the host. Virtually every colonized individual develops chronic active gastritis, yet only ~10–20% ever develop overt clinical disease.

The clinical outcome of infection is not uniform; it branches according to the **anatomical topography of gastritis** and the interaction of **bacterial virulence factors** (chiefly CagA and VacA) with **host genetic susceptibility** (chiefly pro-inflammatory *IL1B*/*IL1RN* polymorphisms). Antral-predominant gastritis drives acid hypersecretion and duodenal ulceration, whereas corpus-predominant or pangastritis drives atrophy, hypochlorhydria, intestinal metaplasia, dysplasia, and ultimately gastric adenocarcinoma (the Correa cascade). *H. pylori* is classified by the IARC as a Group 1 (definite) human carcinogen and is the dominant cause of gastric cancer, with a population attributable fraction averaging ~70% across 27 countries. It also causes gastric MALT (mucosa-associated lymphoid tissue) lymphoma, which regresses completely in most patients after eradication of the bacterium alone.

The disease is diagnosed non-invasively by the urea breath test and monoclonal stool antigen test (serology cannot distinguish active from past infection) and is curable with combination acid-suppression plus antibiotic therapy. Landmark long-term randomized controlled trial

evidence establishes that eradication is effective **primary prevention** of gastric cancer. No licensed vaccine yet exists, although multivalent candidates (VacA/CagA/NAP) have reached human trials. This report synthesizes 15 confirmed findings drawn from 72 papers across all requested domains.

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1. Disease Information

Overview. *H. pylori* infectious disease is chronic colonization of the gastric mucosa by *H. pylori*, a neutralophilic, microaerophilic, spiral (helical) Gram-negative bacterium that produces urease, catalase, and oxidase. The organism was discovered by Marshall and Warren in 1982, linking it to ulcer disease (PMID: 15345211). Colonization causes chronic active type B gastritis in essentially all infected individuals; a minority progress to peptic ulcer disease, gastric adenocarcinoma, or gastric MALT lymphoma (PMID: 9889978).

Key identifiers. - **MONDO:** MONDO:0006781 - **MeSH:** *Helicobacter Infections* (D016481); organism *Helicobacter pylori* - **ICD-10:** B96.81 (*H. pylori* as cause of diseases classified elsewhere); associated gastritis K29.-, peptic ulcer K25–K27 - **ICD-11:** coding under gastritis/duodenitis and infectious agents - **NCBI Taxon:** 210 (*Helicobacter pylori*) - OMIM/Orphanet: not applicable as a primary Mendelian entry — this is an infectious, not a monogenic, disease. Host susceptibility loci such as *IL1B* have OMIM gene entries (see Section 4).

Synonyms / alternative names. *Helicobacter pylori* infection; *H. pylori* infection; *Campylobacter pylori* infection (historical, pre-1989); *Campylobacter pyloridis* infection (original 1983–84 designation); Helicobacter-associated gastritis; type B gastritis (when antral).

Information source. This report derives predominantly from **aggregated disease-level resources** — meta-analyses, systematic reviews, RCTs, consensus guidelines (Maastricht V/VI, ACG), and mechanistic laboratory studies — rather than individual EHR-level patient records.

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2. Etiology

Primary cause — infectious. The disease is caused by infection with *H. pylori*, a necessary cause of the chronic active gastritis underlying all downstream *H. pylori* disease. Core virulence factors are the vacuolating cytotoxin **VacA**, the **cag pathogenicity island / CagA** oncoprotein delivered by a type IV secretion system, **urease**, and **flagellar motility** (PMID: 9889978).

Environmental / lifestyle risk factors. Acquisition is strongly socioeconomic: *"In the vast majority of individuals, infection is acquired during childhood with those of low socioeconomic means and having infected family members being at highest risk for early childhood acquisition"* (PMID: 15758899). Household crowding, an infected family member (especially the mother), and poor sanitation are dominant environmental risk factors. Progression to cancer is further modulated by high dietary salt, smoking, and low ascorbate intake.

Host genetic risk factors. Host pro-inflammatory polymorphisms strongly modify cancer risk. In *H. pylori*-infected Pakistani individuals, combined carriage of *IL-1B*-511*T and *IL-1RN**2 conferred dramatically increased gastric cancer risk (**OR 19.064; 95% CI 2.319–156.7; p = 0.001**): *"A combination of IL-1B-511*T and IL-1RN*2 alleles ... in H. pylori infected individuals had markedly increased risk of GC development"* (PMID: 28110439). In an Algerian cohort, *IL-1β*-31C associated with gastric carcinoma (OR 4.614, 95% CI 1.43–14.81, p = 0.01) and *IL-1β*-511T with chronic atrophic gastritis (OR 4.286, 95% CI 1.54–11.89, p = 0.005) (PMID: 27340011). Interactions among *PGC*, *PTPN11*, *TLR4*, and *IL1B* SNPs further modify carcinogenesis under infection (PMID: 26158864).

Protective factors. Eradication is the principal protective intervention (Sections 11–13). Diets rich in ascorbate and low in salt are associated with reduced progression. No robust protective germline allele has been established with the confidence of the *IL1* risk alleles.

Gene–environment interaction. The paradigm is explicitly gene × environment: the same *IL1* polymorphisms are innocuous without infection but high-risk in its presence. A cumulative *PGC/PTPN11* interaction *"switched from being beneficial to being risky by the status of H. pylori infection"* (PMID: 26158864) — a clean demonstration that host genotype manifests risk only in the infected context.

3. Phenotypes

Most infection is clinically silent. Nearly all colonized individuals develop chronic active gastritis, but the majority remain asymptomatic, and only ~10–20% develop clinical disease over a lifetime (PMID: 15758899). In symptomatic dyspeptic cohorts undergoing endoscopy, **epigastric pain is predominant (67.4–79.2%)**, followed by heartburn (26.1%); gastritis is the most common endoscopic finding (65.5%), with gastric ulcers ~4.6% and carcinoma <1% (PMID: 33251073: *"Epigastric pain was the predominant symptom (79.2%) followed by heartburn (26.1%) ... Gastritis (65.5%) and oesophagitis (33.1%) were the most common findings"*; PMID: 34579657).

Phenotype	Type	HPO suggestion	Frequency / notes
Chronic active gastritis	Histopathologic sign	HP:0031273 (chronic gastritis)	~100% of infected (mostly subclinical)
Epigastric pain	Symptom	HP:0011834	67–79% of symptomatic dyspeptics
Heartburn / dyspepsia	Symptom	HP:0025439 (dyspepsia)	~26%
Peptic (gastric/ duodenal) ulcer	Clinical sign / lesion	HP:0004398 (peptic ulcer)	Minority; PAF for PUD ~57%
Gastric MALT lymphoma	Neoplasm	HP:0012190 (MALT lymphoma)	Rare; PAF ~33% of gastric lymphoma
Gastric adenocarcinoma	Neoplasm	HP:0012126 (gastric cancer)	PAF ~70% (non-cardia)
Iron-deficiency anemia	Lab abnormality	HP:0001891	Extragastric association
Vitamin B12 deficiency	Lab abnormality	HP:0100502	Extragastric association
Immune thrombocytopenia	Lab/clinical sign	HP:0001973 (thrombocytopenia)	Extragastric association

Characteristics. Colonization onset is in childhood, but clinical disease typically manifests decades later. Children may nonetheless develop chronic active gastritis, ulcers, and MALT lymphoma (PMID: 15758899). Severity ranges from mild asymptomatic gastritis to lethal carcinoma; progression is chronic and, for the neoplastic branch, stepwise over decades.

Quality-of-life impact. Symptomatic dyspepsia and peptic ulcer disease impair daily functioning; gastric cancer and lymphoma carry the greatest burden. Disease burden extends beyond cancer: attributable fractions were 57% for peptic ulcer disease, 7% for dyspepsia, and 33% for gastric lymphoma (PMID: 41236450).

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4. Genetic / Molecular Information

This is an infectious disease, so there are **no human causal germline genes**. The relevant genetics are (a) host susceptibility modifier loci and (b) bacterial virulence and resistance genes.

Host modifier genes. - ***IL1B*** (interleukin-1β; HGNC:5992) — promoter polymorphisms -511 C/T (rs16944), -31 (rs1143627) and rs1143627 raise gastric cancer / atrophic gastritis risk under infection (PMID: 28110439, PMID: 27340011). - ***IL1RN*** (IL-1 receptor antagonist; HGNC:6000) — the *2 VNTR allele synergizes with *IL1B* (PMID: 28110439). - ***PGC*, *PTPN11*, *TLR4** — *interacting susceptibility loci (PMID: 26158864).* - **CYP2C19*** — pharmacogenetic modifier of treatment: rapid/ultrarapid metabolizers have 2.52-fold higher eradication failure with PPI-based regimens (PMID: 34358488).

Bacterial virulence and resistance genes (the true "pathogenic variants" of this disease). - ***cagA*** and the *cag* pathogenicity island / type IV secretion system — principal oncogenic determinant (PMID: 17237808, PMID: 28338646). - ***vacA*** — vacuolating cytotoxin, allelic mosaicism (s1/m1 more virulent). - ***ureA/ureB/ureI*** — urease and its pH-gated urea channel (colonization). - **Resistance mutations:** 23S rRNA A2142G/A2143G (clarithromycin), *rdxA* (metronidazole), *pbp1A* (amoxicillin), *gyrA/gyrB* (levofloxacin) (PMID: 41966313).

Functional consequence. CagA acts as a **bacterial oncoprotein** with gain-of-function signaling in host cells (Section 6). Host *IL1B* risk alleles are gain-of-function for IL-1β expression, amplifying gastric inflammation and acid suppression.

Epigenetics. *H. pylori*-driven chronic inflammation induces aberrant promoter DNA methylation in gastric epithelium, contributing to the field of cancerization. CagA also transactivates *CDX1* and induces *MUC2*, marking intestinal transdifferentiation (PMID: 17237808).

Chromosomal abnormalities. Not applicable at the germline level; somatic copy-number and mutational changes accumulate in gastric epithelium during the cancer cascade.

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5. Environmental Information

Environmental / lifestyle factors. Low socioeconomic status, crowding, poor sanitation, and contaminated water in childhood are the principal drivers of acquisition (PMID: 15758899, PMID: 11488660). High dietary salt, smoking, and low vitamin C promote progression along the cancer cascade.

Infectious agent. The causative pathogen is *H. pylori* (NCBI:txid210), a Gram-negative, microaerophilic, spiral bacterium. Related non-pylori gastric *Helicobacter* species (*H. suis*, *H. heilmannii*, *H. felis*, *H. salomonis*, *H. bizzozeronii*) cause a smaller fraction of human gastric disease and are zoonotic (Section 14) (PMID: 26155631).

CHEBI-relevant chemical entities: urea (CHEBI:16199), ammonia (CHEBI:16134), nitrite (CHEBI:16301), ascorbic acid (CHEBI:29073), nickel ion (CHEBI:49786, urease cofactor).

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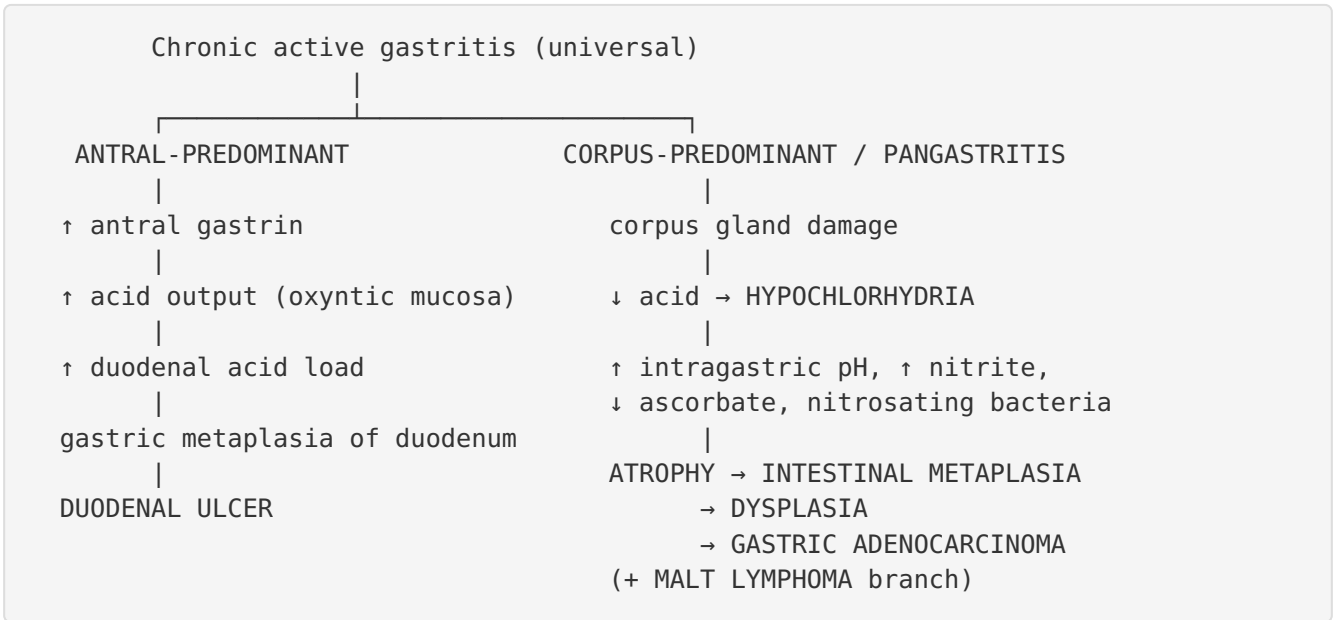
6. Mechanism / Pathophysiology

Ordered causal chain

1. **Ingestion** of *H. pylori* (oral–oral / fecal–oral, in childhood) **leads to** arrival in the gastric lumen.
2. **Urease-mediated acid resistance** — constitutive Ni^{2+}-dependent urease hydrolyzes urea to ammonia; the pH-gated urea channel UreI opens when periplasmic pH falls below ~6.2 — **allows** transient survival in gastric acid. *"Acute acid resistance depends on the constitutive synthesis of a neutral pH optimum urease ... Gastric juice urea is able to rapidly access intrabacterial urease when the periplasmic pH falls below approximately 6.2 owing to pH-gating of a urea channel, UreI"* (PMID: 12471160).

3. **Flagellar motility + adhesins enable** penetration of the mucus layer and colonization of the epithelial-surface niche, avoiding mucus clearance (PMID: 24914320).
4. **Persistent colonization induces** chronic active gastritis (neutrophil + lymphocyte infiltration) (PMID: 9889978).
5. **CagA injection via T4SS** → tyrosine phosphorylation by Src-family kinases → **binds and activates SHP-2 phosphatase**, causing "hummingbird" cell-morphological transformation: *"Tyrosine-phosphorylated CagA specifically binds to and activates SHP-2 phosphatase, thereby inducing cell-morphological transformation"* (PMID: 17237808, PMID: 28338646).
6. **Phosphorylation-independent CagA → binds E-cadherin**, disrupting the E-cadherin/β-catenin complex → **cytoplasmic/nuclear β-catenin accumulation** → transactivates *CDX1* → induces goblet-cell marker *MUC2* → **intestinal metaplasia**: *"The CagA/E-cadherin interaction impairs the complex formation between E-cadherin and beta-catenin, causing cytoplasmic and nuclear accumulation of beta-catenin. CagA-deregulated beta-catenin then transactivates beta-catenin-dependent genes such as cdx1"* (PMID: 17237808).
7. **Chronic inflammation** → dicarbonyl/isolevuglandin electrophiles form DNA adducts → **DNA damage and increased somatic mutation frequency** (PMID: 33189701).

Branch point — topography of gastritis determines outcome (PMID: 9494857: *"Subjects in whom the infection produces acid hypersecretion develop duodenal ulcer disease ... In subjects in whom the infection induces marked hypochlorhydria, there is an increased risk of gastric cancer"*; PMID: 9376611):



Molecular pathways / cellular processes. CagA-driven signaling co-opts SHP-2/ERK-MAPK, NF-κB, and Wnt/β-catenin pathways. Chronic inflammation, apoptosis dysregulation, and oxidative/lipid-peroxidation DNA damage are the core cellular processes. Relevant GO biological-process terms: **GO:0006954** (inflammatory response), **GO:0016477** (cell migration, "hummingbird"), **GO:0006281** (DNA repair). CL cell types: **CL:0000584** (enterocyte, metaplastic), **CL:0002180** (mucous cell of stomach), **CL:0000236** (B cell, MALT lymphoma).

Immune involvement. Chronic Th1/Th17-skewed gastric inflammation is central both to pathology and to the immunity candidate vaccines aim to elicit (PMID: 37971157, PMID: 40776465).

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7. Anatomical Structures Affected

- **Primary organ:** stomach (**UBERON:0000945**), specifically the gastric mucosa. Antrum (**UBERON:0001165**) in the duodenal-ulcer branch; corpus/body (**UBERON:0001161**) in the cancer branch.
- **Secondary sites:** duodenum (**UBERON:0002114**) — gastric metaplasia and duodenal ulcer; systemic/hematologic involvement in extragastric disease.
- **Body system:** digestive system (**UBERON:0001007**).
- **Tissue level:** gastric surface mucous epithelium; gastric glands (parietal, chief, G-cells, D-cells, ECL cells); mucosa-associated lymphoid tissue for MALT lymphoma.
- **Cell populations (CL):** gastric surface mucous cell (CL:0002180), parietal cell (CL:0000162), chief cell (CL:0000155), G-cell/gastrin cell, ECL cell, gastric stem/progenitor cells, mucosal B lymphocytes (CL:0000236).
- **Subcellular (GO cellular component):** bacterial urease is cytoplasmic/periplasmic; CagA localizes to the host inner plasma membrane (**GO:0005886**) and translocates to the nucleus (**GO:0005634**), where β-catenin accumulates.
- **Localization / lateralization:** not applicable — a midline, non-lateralized luminal organ; disease is defined by mucosal topography (antrum vs corpus) rather than laterality.

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8. Temporal Development

- **Onset:** colonization in early childhood; insidious and chronic. If untreated, *"H pylori infection is lifelong"* (PMID: 15758899).
- **Progression:** clinical disease typically manifests decades after acquisition. The neoplastic branch follows the multi-decade **Correa cascade**: chronic gastritis → atrophy → intestinal metaplasia → dysplasia → adenocarcinoma. Clarithromycin resistance rises with disease severity along the cascade: CNAG 33.3% → CAG 31.7% → gastric ulcer 59.3% → MALT-L 54.1% → gastric cancer 69.6% (PMID: 42365653).
- **Course:** chronic and progressive if untreated; ulcer disease can be episodic/relapsing-remitting.
- **Remission / critical windows:** eradication is treatment-induced remission. Earlier eradication (before premalignant lesions develop) yields the greatest cancer-prevention benefit — a critical intervention window (Section 11).

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9. Inheritance and Population

Epidemiology. Global population prevalence declined modestly *"from 43.1% (95% credible interval 39.1-47.4) in 1990 to 36.3% (33.2-39.6) in 2023"* (PMID: 42757050). Prevalence approaches 100% in some developing regions and is falling in high-income countries. The organism is the dominant cause of gastric cancer: the population attributable fraction averaged **69.7%** across 27 countries (*"ranging from 40.7% in Malaysia to 82.3% in South Africa"*; risk ratio 5.9) (PMID: 41115989).

Inheritance. Not applicable — this is an acquired infection, not a heritable Mendelian disorder. Host **susceptibility is polygenic/multifactorial** (e.g., *IL1B*, *IL1RN*, *TLR4*). Familial clustering reflects **shared household transmission** (chiefly mother-to-child), not genetic inheritance.

Population demographics. Higher prevalence in low-socioeconomic and developing-country populations; East Asian, Latin American, and some African populations bear the highest gastric-cancer burden, partly attributable to more virulent East-Asian *cagA* genotypes. Acquisition is concentrated in childhood; prevalence rises with age in cross-section (cohort effect). Infection sex ratio is roughly equal; gastric cancer shows a male predominance (~2:1).

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10. Diagnostics

Non-invasive tests. Pooled sensitivity/specificity from meta-analysis (PMID: 38143438: *"The pooled sensitivity and specificity for HpSA were 72.5 and 94.7%, for Serology 83.7 and 73.3%, and for UBT 96.4 and 88.3%"*):

Test	Sensitivity	Specificity	Diagnostic OR	Note
Urea breath test (UBT)	96.4%	88.3%	94.5	Best overall; detects active infection
Monoclonal stool antigen (HpSA/SAT)	72.5%	94.7%	47.9	High specificity; reliable, non-invasive
Serology (IgG)	83.7%	73.3%	14.2	Cannot confirm active infection

Serology *"is useful only in selected settings because it cannot reliably distinguish current from previous infection"* (PMID: 42724990). UBT sensitivity falls in advanced atrophic/metaplastic mucosa; qPCR and monoclonal SAT show superior performance across disease stages, and multimodal strategies (e.g., SAT + histology) achieve the highest sensitivity (PMID: 41985696).

Invasive / reference-standard tests. Endoscopic biopsy with histology (H&E, IHC), rapid urease test (CLO test), and culture remain reference standards and permit resistance testing and staging of premalignant lesions (PMID: 42724990). Molecular detection of 23S rRNA resistance mutations guides tailored therapy.

Genetic/omics diagnostics. Human germline genetic testing is not diagnostically indicated. Bacterial molecular testing (qPCR, sequencing for resistance mutations) is increasingly used; *CYP2C19* genotyping can inform PPI dosing.

Differential diagnosis. Functional dyspepsia, NSAID-induced ulcer, autoimmune (type A) gastritis, Zollinger-Ellison syndrome, gastroparesis, biliary disease, and gastric neoplasia of other etiology.

Screening. Population "test-and-treat" / "screen-and-treat" programs are recommended in high-incidence regions and are being trialed in adolescents (e.g., Chile, NCT05926804) (PMID: 39890135).

11. Outcome / Prognosis

Untreated course. Lifelong infection; ~10–20% develop clinical disease. Most morbidity/mortality flows from gastric adenocarcinoma (poor prognosis when diagnosed late) and peptic ulcer complications (bleeding, perforation).

With eradication. Excellent — eradication cures gastritis, heals ulcers, prevents recurrence, and reduces cancer risk. **MALT lymphoma prognosis is outstanding:** in a 398-patient multicenter cohort, overall complete remission (CR) was **97.0%** (*"98.3% in HPP and 93.3% in HPN (P = 0.018)"*), 91.2% presented at localized stage IE, and relapse was only 8.3% over a median 179 weeks (PMID: 41776408). Pediatric cases also regress completely after eradication alone (PMID: 42034998).

Cancer prevention (landmark RCT). In 1,630 asymptomatic infected individuals in high-risk southern China with **26.5 years of follow-up**, eradication reduced gastric cancer incidence (2.57% vs 4.31%; **HR 0.57, 95% CI 0.33–0.98**), with greater benefit in those without premalignant lesions (**HR 0.37, 95% CI 0.15–0.95**) (PMID: 35364066). A Cochrane meta-analysis confirmed eradication reduces both gastric cancer incidence (RR 0.54, 95% CI 0.40–0.72) and gastric-cancer mortality (RR 0.61, 95% CI 0.40–0.92) (PMID: 32628791).

Prognostic factors. Presence of premalignant lesions at treatment predicts residual cancer risk; *cagA*-positive/East-Asian strains and virulent *vacA* alleles and high-risk host genotypes worsen prognosis; successful eradication vs persistent infection is itself prognostic (HR 0.46) (PMID: 35364066).

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12. Treatment

First-line pharmacotherapy (Maastricht V/VI and ACG guidelines): - **Clarithromycin-based triple therapy** (PPI + clarithromycin + amoxicillin) in regions with clarithromycin resistance <15% (PMID: 29600697). - **Bismuth quadruple therapy** (PPI + bismuth + tetracycline + metronidazole) in high-resistance regions — *"Greater than 90% eradication success can be obtained by bismuth-containing quadruple therapy"* (PMID: 26808355). - **Potassium-competitive acid blockers (vonoprazan)** increasingly replace PPIs, giving more potent acid suppression and higher eradication rates (PMID: 29600697, PMID: 42410737).

Rescue / second-line: levofloxacin-based triple therapy, rifabutin- or furazolidone-based regimens, or susceptibility-guided tailored therapy (PMID: 29600697).

Antibiotic resistance is the key driver of failure: *"Clarithromycin resistance in H. pylori is mainly driven by A2142G/A2143G mutations (19-34%), while metronidazole resistance exceeds 84% in East Asia via rdxA mutations. Amoxicillin resistance (6-34%) links to pbp1A variants, and levofloxacin resistance (31-66%) to gyrA/B mutations"* (PMID: 41966313).

Pharmacogenomics. Host **CYP2C19** rapid/ultrarapid-metabolizer phenotype raises PPI-based eradication failure — *"enhanced vs poor metabolizer phenotypes were associated with a 2.52-fold significantly higher likelihood of eradication failure"* (PMID: 34358488) — the rationale for genotype-guided PPI dosing or P-CABs, which are less CYP2C19-dependent.

NCIT term suggestions: Proton Pump Inhibitor (NCIT:C1454), Clarithromycin (NCIT:C349), Amoxicillin (NCIT:C284), Metronidazole (NCIT:C619), Tetracycline (NCIT:C859), Bismuth Subsalicylate, Levofloxacin (NCIT:C1615), Vonoprazan.

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13. Prevention

- **Primary prevention:** *H. pylori* eradication is proven primary prevention of gastric cancer (PMID: 35364066; Cochrane PMID: 32628791). Improved sanitation, clean water, and reduced crowding lower transmission.
- **Secondary prevention:** test-and-treat / screen-and-treat programs; endoscopic surveillance of premalignant lesions (PMID: 39890135).
- **Tertiary prevention:** eradication in patients with ulcers, MALT lymphoma, or post-resection gastric cancer to prevent recurrence.
- **Vaccination (not yet available):** *"notwithstanding a proof-of-concept that vaccination can protect children from acquisition of H. pylori infection, there are currently no advanced vaccine candidates with only a single vaccine in Phase I clinical trial"* (PMID: 29627231). A parenteral three-antigen vaccine (**VacA** + **CagA** + **NAP**) *"was well tolerated and highly immunogenic in healthy adults"* in a randomized phase 1/2 human challenge study (PMID: 30042064). Leading antigens: UreB, VacA, CagA, HspA; effective immunity likely requires Th1/Th17 responses (PMID: 37971157, PMID: 40776465).

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14. Other Species / Natural Disease

- **Taxonomy:** *H. pylori* (NCBI:txid210). Non-pylori gastric *Helicobacter* species that infect humans zoonotically include *H. suis*, *H. felis*, *H. salomonis*, *H. bizzozeronii*, and *"H. heilmannii"*.
- **Natural / zoonotic disease:** *"Animal Helicobacter species other than Helicobacter pylori are also able to cause human gastritis, gastric ulcers, and MALT lymphomas"* (PMID: 26155631). *H. suis "infects the stomach of humans and animals such as dogs, cats, pigs, and rodents, the latter giving rise to zoonotic infection"* (PMID: 28534071).
- **Reservoirs / host range:** dogs, cats, pigs, primates, cattle, rodents, and even rabbits harbor gastric *Helicobacter* species (PMID: 15751578). In humans with non-*H. pylori* spiral bacteria, PCR identified *Candidatus* H. suis (30.9%), *H. salomonis* (11.4%), and *H. felis* (8.9%), with dogs and cats as reservoirs (PMID: 15872252).
- **Comparative biology:** these species share cholesteryl-α-glucoside cell-wall biology with *H. pylori* (56% αCgT amino-acid identity) (PMID: 28534071), indicating conserved gastric-colonization mechanisms.

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15. Model Organisms

Wild-type rodents do not spontaneously develop gastric adenocarcinoma, but it can be induced by *Helicobacter* infection (PMID: 15080846):

Model	Type	Phenotype recapitulation	Application
Mongolian gerbil (*Meriones unguiculatus*)	Induced infection	Develops adenocarcinoma after *H. pylori* infection; recapitulates atrophy→metaplasia→dysplasia→carcinoma	Virulence-gene, dietary-salt, carcinogenesis studies
C57BL/6 mouse + *H. felis*	Induced infection	Gastric cancer after long-term infection	Host-immune / inflammation studies
INS-GAS transgenic mouse (FVB/N insulin-gastrin)	Genetic (hypergastrinemic)	Spontaneous carcinoma in old age, greatly accelerated by *Helicobacter*	Gastrin/acid–cancer axis, chemoprevention

Key quotes: *"gastric adenocarcinoma can be induced in Mongolian gerbils by H. pylori infection and in C57BL/6 mice by long-term H. felis infection"* and *"the hypergastrinaemic INS-GAS mouse, in which gastric carcinoma develops spontaneously in old animals, but the process is greatly accelerated by Helicobacter infection"* (PMID: 15080846). These models linked carcinogenesis to a specific molecular mechanism: *"EtHOBA inhibited gastric carcinoma in infected INS-GAS mice and gerbils and attenuated isoLG adducts, DNA damage, and somatic mutation frequency"* (PMID: 33189701).

Limitations: rodent gastric anatomy/physiology differs from human; models often require East-Asian or highly virulent strains and long timelines; the MALT-lymphoma phenotype is imperfectly modeled.

Mechanistic Model / Interpretation

The unifying model is that a single infectious agent produces a **spectrum of divergent diseases** governed by three axes: (1) **anatomical topography** of the induced gastritis, (2) **bacterial virulence genotype** (*cagA*, *vacA*), and (3) **host inflammatory genotype** (*IL1B*/*IL1RN*). Colonization is universally possible because urease/UreI acid resistance plus motility and adhesion let the organism occupy the gastric mucus niche. Once resident, the outcome bifurcates:

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INFECTION (childhood) ──> urease/motility/adhesion ──> chronic active gastritis
      │
      ├─ antral-predominant + acid hypersecretion ──> DUODENAL ULCER
      │
      └─ corpus/pangastritis (amplified by IL1B risk alleles + CagA)
            │  hypochlorhydria, ↑nitrite, ↓ascorbate
            │  CagA→SHP-2 (proliferation) + CagA→E-cadherin/β-catenin→CDX1/MUC2 (metaplasia)
            │  inflammation→isoLG DNA adducts→somatic mutation
            └─> ATROPHY→METAPLASIA→DYSPLASIA→ADENOCARCINOMA  (+ MALT LYMPHOMA)
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The therapeutic corollary is powerful: because the bacterium is a *necessary* upstream cause, **removing it reverses or prevents downstream disease** — ulcers heal, MALT lymphoma regresses in ~97%, and cancer incidence falls by ~40–60% — with the largest benefit when eradication precedes irreversible premalignant change.

Evidence Base

PMID	Contribution	Role
42757050	Global prevalence 36.3% (2023)	Epidemiology
41115989	Gastric-cancer PAF ~70%	Causality/burden
41236450	PAF for PUD/dyspepsia/ lymphoma	Burden beyond cancer
28110439 / 27340011	*IL1B*/*IL1RN* gene×infection risk	Host genetics
26158864	SNP interaction effect-modification	Host genetics
17237808 / 28338646	CagA→SHP-2 and β-catenin/ CDX1	Mechanism (in vitro)
12471160 / 24914320	Urease/UreI, mucus-niche colonization	Colonization biology
9494857 / 9376611	Topography branch point	Pathophysiology
33189701	isoLG DNA adducts, EtHOBA	DNA-damage mechanism (model)
38143438 / 42724990 / 41985696	UBT/SAT/serology accuracy	Diagnostics
29600697 / 26808355 / 41966313 / 34358488	Regimens, resistance, CYP2C19	Treatment
35364066 / 32628791	RCT/Cochrane cancer prevention	Prevention/prognosis
41776408 / 42034998	MALT lymphoma regression	Prognosis
29627231 / 30042064 / 37971157	Vaccine status/candidates	Prevention
15080846	Gerbil/INS-GAS models	Model organisms
26155631 / 28534071 / 15872252 / 15751578	Zoonotic non-pylori species	Comparative biology
15758899 / 11488660 / 15345211	Transmission, childhood acquisition	Etiology/epidemiology

Evidence source types span **human clinical** (RCTs, cohorts, meta-analyses, guidelines), **model organism** (gerbil, mouse), and **in vitro** (CagA signaling, urease biochemistry).

Limitations and Knowledge Gaps

1. **Extragastric associations** (iron-deficiency anemia, B12 deficiency, ITP) rest partly on association and guideline consensus rather than mechanistic proof; molecular-mimicry mechanisms for ITP remain hypothesized (PMID: 42410737, PMID: 28727656).
2. **Prevalence heterogeneity** is large; global averages mask 5-fold regional variation, and declining-prevalence trends may not generalize.
3. **Host genetic risk estimates** (e.g., *IL1B* OR up to 19) come from modest single-population studies with wide confidence intervals and require multi-ethnic replication.
4. **No licensed vaccine** exists; correlates of protective immunity are incompletely defined.
5. **Antibiotic resistance** is rising and geographically variable, undermining empiric regimens; real-world eradication rates increasingly fall below the >90% target.
6. **Non-pylori Helicobacter disease** is under-recognized because standard tests target *H. pylori*.
7. **Model organisms** imperfectly recapitulate human MALT lymphoma and the full Correa cascade timeline.

Proposed Follow-up Experiments / Actions

1. **Genotype-guided therapy trials:** prospectively test *CYP2C19*- and 23S rRNA-resistance-guided regimens (P-CAB-based) versus empiric therapy for eradication success.
2. **Multi-ethnic host-genetics replication:** validate *IL1B*/*IL1RN*/*TLR4* risk in large, diverse cohorts with standardized cancer endpoints.
3. **Vaccine development:** advance multivalent VacA/CagA/NAP/UreB candidates through controlled human infection models with defined Th17 correlates.
4. **Screen-and-treat outcome studies:** complete adolescent screen-and-treat RCTs (e.g., NCT05926804) to quantify population-level cancer prevention and cost-effectiveness.

5. **Chemoprevention:** translate dicarbonyl-scavenger (EtHOBA-class) findings from gerbil/INS-GAS models toward human premalignant-lesion trials.
6. **Diagnostics in advanced mucosa:** deploy qPCR/monoclonal-SAT multimodal panels for reliable detection in atrophic/metaplastic stomachs where UBT underperforms.
7. **Non-pylori Helicobacter surveillance:** implement broad-range Helicobacter PCR in culture-negative gastritis with animal contact to capture zoonotic cases.

Report compiled from 15 confirmed findings across 72 reviewed papers over 5 investigation iterations. Ontology suggestions: MONDO:0006781; NCBI:txid210; UBERON:0000945/0001165/0001161/0002114; CL:0002180/0000162/0000155/0000236; GO:0006954/0016477/0006281/0005886/0005634; CHEBI:16199/16134/16301/29073/49786; HPO terms per Section 3; NCIT drug terms per Section 12.
